DS4I Assignment 2: Neural Networks

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Introduction

Introduction

Research context

- Handwriting varies across individuals and across writing sessions
- Despite this variability, individuals may retain distinctive handwriting characteristics
- The study investigates whether neural networks can identify writers from handwritten digit images

Research questions

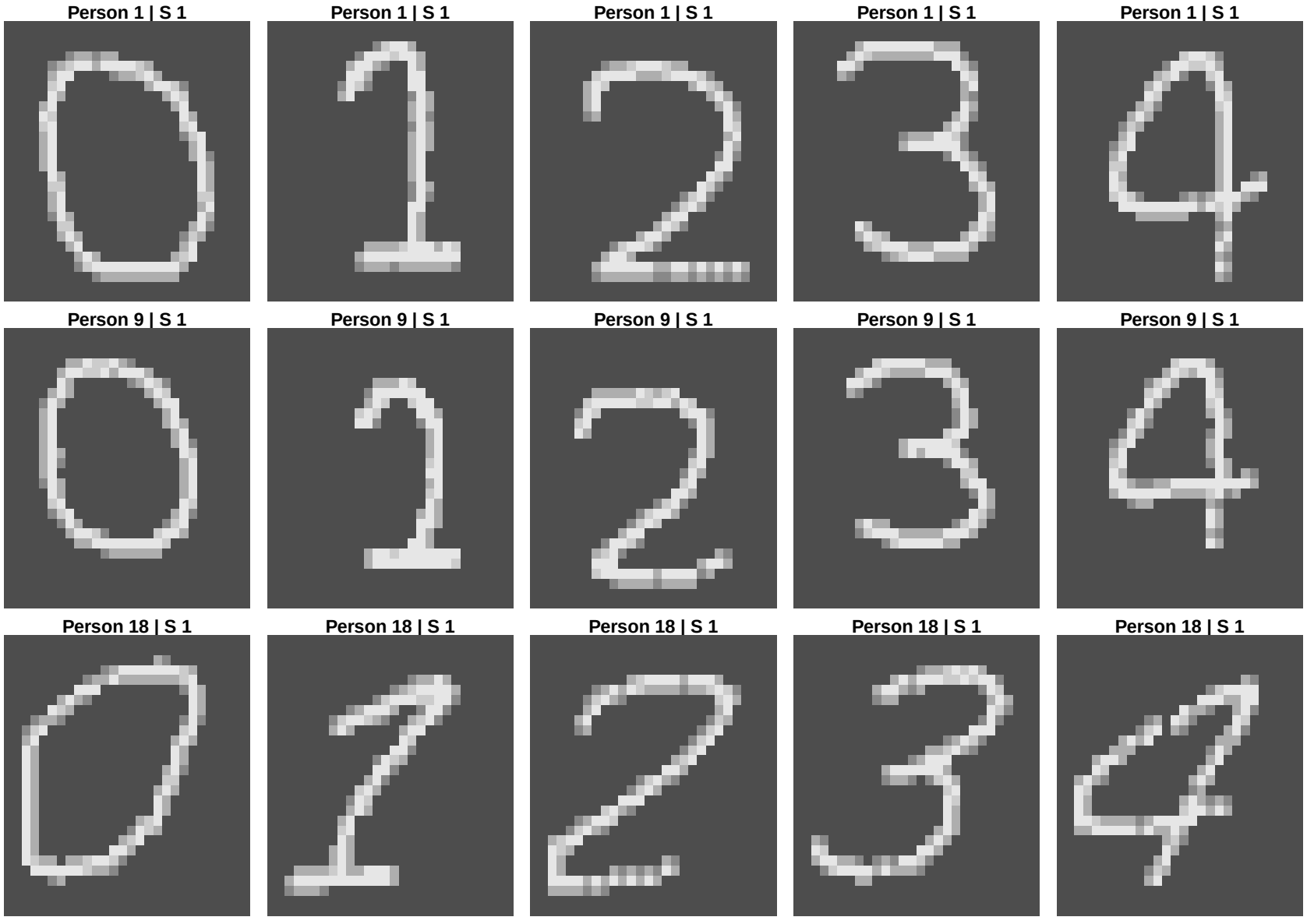
- Can a writer be identified by a single handwritten digit?
- Can a model determine whether two handwritten digits belong to the same writer?

Approach

- Convolutional Neural Network
- Siamese Neural Network

Exploratory Data Analysis

Visual findings



Visual findings

- Different writers display visible handwriting characteristics that could help a model identify them
- Handwriting varies within the same person, but certain individual characteristics persist across sessions
- A writer's style can be expressed across different digits, potentially helping a model recognise a person even when the digit changes

Variance decomposition

Percentage of variance explained

Table 1: Table. Percentage of Variance Decomposition for Features

	Digit	Person ID	Session	Person ID : Session	Residuals
h	7.6	62.9	0.1	2.9	26.5
w	43.5	40.5	0.4	6.8	8.8
cx	31.6	34.4	0.6	10.3	23.1
cy	15.0	10.3	0.4	17.0	57.3

- Digit dimensions, particularly height, are more stable indicators of writer identity than vertical placement

Euclidean distance

Table 2: Table. Euclidean Distances and Ratios Across Digits (0 to 4)

	Digit 0	Digit 1	Digit 2	Digit 3	Digit 4
same_writer_diff_session	7.06	6.48	6.58	6.65	6.35
diff_writer_same_session	8.78	7.85	8.09	8.17	7.98
diff_writer_diff_session	8.79	7.85	8.10	8.18	7.98
Ratio (Same/Diff Writer)	0.80	0.83	0.81	0.81	0.80

Euclidean distance

- There is a measurable signal for distinguishing writers, but handwriting changes between sessions enough to make identification challenging
- The similarity of distances for different writers within and across sessions also supports the finding that there is little overall session effect

Convolutional Neural Network

Methods

Question and task

- Question: can a writer be identified from a single handwritten digit?
- Input: one 28×28 image
- Output: a probability for each of the 20 writers (softmax); the predicted writer has the highest probability
- Each image is labelled with its writer, whatever digit it shows; the network is not told the digit
- Split by session: train sessions 1 – 3, validation session 4, test session 5

Architecture and calibration

- CNN: two convolutional layers (16 filters, 3×3), dropout, 2×2 max pooling, softmax output
- Linear softmax: pixels straight to the 20 writers. Nearest centroid: average image per writer and digit

Hyperparameter	Default CNN	Tuned CNN	Linear softmax	Nearest centroid
Hidden layers	2 conv.	2 conv.	none	none
Activation	ReLU	tanh	none	none
Dropout rate	0.2	0.7	none	none
Learning rate	1×10^{-2}	1×10^{-4}	1×10^{-4}	none
L2 regularisation	none	0.1	none	none
Epochs	10	10	20	none
Told the digit	no	no	no	yes

Tuning

- Stage 1: activation function, learning rate, dropout rate (72 combinations)
- Stage 2: number of filters, kernel size, pool size (12 combinations)
- Stage 3: number of epochs
- Stage 4: L2 penalty ν (8 values)
- Validation session only (session 4)
- Average over three random seeds reported

Metrics

Accuracy

- Proportion of images assigned to the correct writer (correct predictions divided by all images)

Cross-entropy

- Quality of the predicted probabilities (lower is better; guessing evenly gives 3.0)

$$\text{Cross-entropy} = -\frac{1}{n}\sum_{i=1}^{n}\log \hat{p}_i$$

where $\hat{p}_i$ is the probability given to the true writer of image i

Choosing between candidates (stages 2 and 3)

- Keep those within 30% of the lowest cross-entropy: $\text{CE} \leq 1.3 \times \text{CE}_{\min}$
- Of those, choose the highest validation accuracy

where $\hat{p}_i$ is the probability given to the true writer of image i

Results

Tuning the L2 penalty

ν	Training	Validation	Cross-entropy
0	0.979	0.591	1.892
0.003	0.979	0.593	1.727
0.01	0.975	0.600	1.529
0.03	0.957	0.606	1.392
0.1	0.862	0.612	1.447
0.3	0.165	0.135	2.986

- Training and validation accuracy; cross-entropy on the validation session (mean of three seeds)
- Dark yellow: value used ($\nu = 0.1$); light yellow: lowest cross-entropy; red: the model cannot learn

Training versus validation

Model	Training	Validation	Gap
Default CNN	99.4%	44.8%	55 points
Tuned CNN	86.0%	60.4%	26 points

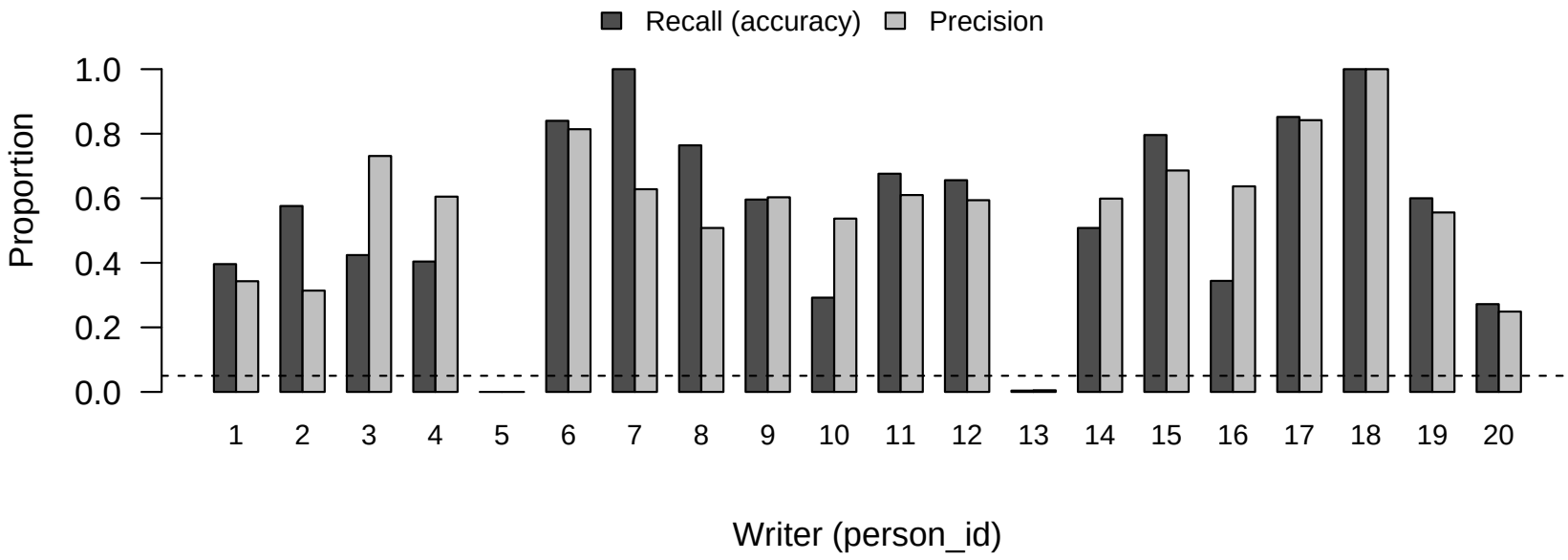
- The default CNN memorises the training sessions
- Tuning halves the gap, but a large gap remains
- The tuned score is optimistic: its settings were chosen on session 4

Test performance (session 5)

Model	Accuracy	Cross-entropy
Nearest centroid	47.3%	–
Linear softmax	43.6%	–
Default CNN	49.3%	5.87
Tuned CNN (final)	**55.0%**	**1.39**
Guessing among 20 writers	5.0%	3.00

- 95% interval for the tuned CNN: 42.7% to 67.1%
- Versus the linear model: $+11.4$ points (95% interval 4.1 to 18.9)
- Versus the default CNN: $+5.7$ points (interval -1.9 to 13.6), not distinguishable from zero

Which writers, and where the errors go



- Writers 7 and 18: all correct. Writer 5: none. Writer 13: 1 of 250
- Writer 10 $\rightarrow$ 2: 71% of images; writer 5 $\rightarrow$ 13: 60%
- Writer 2 received 459 predictions, only 31% correct (fair share: 250)

Conclusions

- A writer can be identified well above chance (55% against 5%), but not reliably
- Likely reason: a writer's digits change between sessions by about 80% of the distance to another writer's
- Limits: one validation and one test session, one training run per model, simulated data, a closed set of 20 writers

Siamese Neural Network

Methods

General Siamese structure

- Input: 2 images
- Uses a CNN
- Output: distance between embeddings of images
- Distance infers dissimilarity
- Contrastive loss:

$$L = yD^2 + (1-y)max(0, m-D)^2$$

Input pairs

Training:

Validation:

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session
- different writers, different sessions

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session
- different writers, different sessions
- sessions 1 - 5

Metrics

Sensitivity

- Proportion of actual positive cases correctly identified

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Specificity

- Proportion of actual negative cases correctly identified

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

Balanced accuracy

- Accounts for imbalanced classes

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}$$

Default calibration

Hyperparameter	Value
Hidden layers	2
Activation functions	ReLU
Number of filters	16 and 2×16
Filter size	3×3
Learning rate	1×10^{-3}
Dropout rate	0.2
L2 regularisation	1×10^{-3}
Layer 1 pooling window	2×2
Embedding length	16
Epochs	15

Tuning

- Hidden layers
- Activation functions
- Filter (square) size
- Number of filters
- **Average validation** performance metrics reported

Results

Tuning hidden layers

Number of Hidden Layers	Contrastive Loss	Balanced Accuracy
1	0.214	0.773
2	0.105	0.966

Tuning activation functions

Activation Functions	Contrastive Loss	Balanced Accuracy
relu_relu	0.105	0.966
tanh_relu	0.103	0.971
relu_tanh	0.108	0.982
tanh_tanh	0.113	0.980

Tuning filter numbers and size

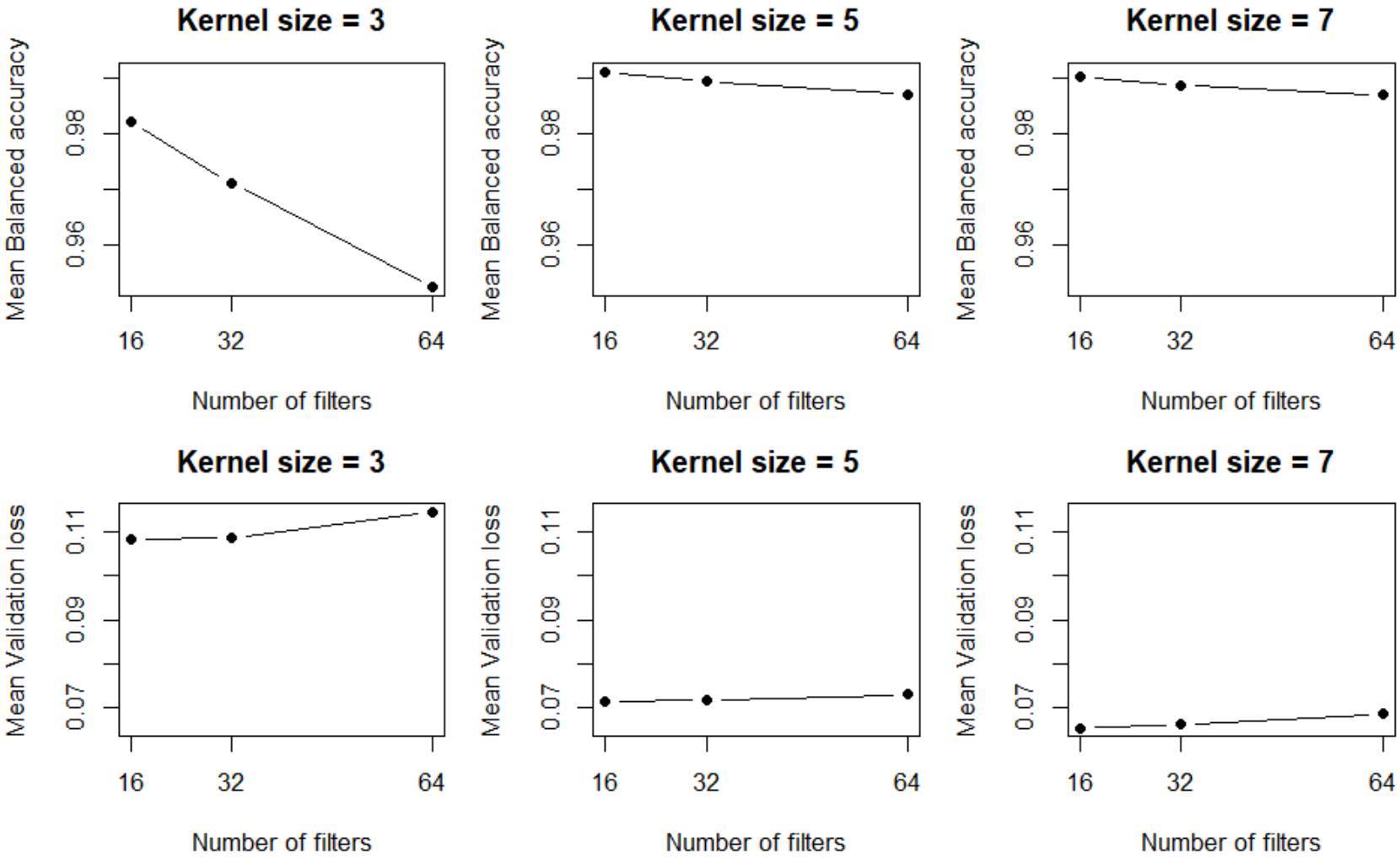


Figure 1

Final model

Tuned inputs

- Hidden layers: 2
- Activation functions: reLU and tanh
- Filter (square) size: 5×5
- Number of filters: 16 and 32
- Epochs: 6

Outputs

- Test Balanced accuracy: 0.751
- Test Loss: 0.104
- Distance threshold: 0.247

Session results

Session Type	n	Sensitivity	Specificity	Balanced Accuracy
Same	2000	0.899	0.995	0.947
Different	2000	0.112	0.997	0.554

Digit results

Digit	Sensitivity	Specificity	Balanced Accuracy
0	0.501	0.995	0.748
1	0.532	0.995	0.763
2	0.513	0.998	0.755
3	0.476	1.000	0.738
4	0.510	0.992	0.751

Writer results (1 - 10)

Writer	Sensitivity	Specificity	Balanced Accuracy
1	0.538	0.991	0.765
2	0.432	1.000	0.716
3	0.481	1.000	0.740
4	0.518	1.000	0.759
5	0.476	0.988	0.732
6	0.444	0.990	0.717
7	0.587	1.000	0.793
8	0.429	0.991	0.710
9	0.523	1.000	0.761
10	0.557	1.000	0.778

Writer results (11 - 20)

Writer	Sensitivity	Specificity	Balanced Accuracy
11	0.495	1.000	0.748
12	0.520	1.000	0.760
13	0.403	0.991	0.697
14	0.453	0.983	0.718
15	0.484	1.000	0.742
16	0.543	1.000	0.772
17	0.542	0.987	0.765
18	0.613	1.000	0.806
19	0.557	1.000	0.778
20	0.552	1.000	0.776

Discussion

CNN accuracy

- Achieved 55.0% writer identification accuracy, well above the 5% random-guessing baseline
- Outperformed the linear baseline by 11.4 percentage points
- Similar accuracy to the default CNN, but substantially better probability calibration (cross-entropy: 1.39 vs. 5.87)
- Test accuracy had a wide confidence interval of 42.7%–67.1%, indicating uncertainty

CNN model limitations

- Accuracy varied significantly between writers, ranging from 0% to 100%
- Misclassifications were systematic, with similar-looking digits often assigned to the same incorrect writer
- Cross-session handwriting variation made a writer's new samples resemble another writer's training samples
- Tuning reduced the training–validation accuracy gap from 55 to 26 percentage points, but overfitting remained
- The test accuracy of 55.0% is more reliable than the 60.4% validation accuracy

Siamese accuracy

- Performed well when identifying pairs of handwriting samples from the same session
- Performance declined substantially when comparing handwriting across different sessions
- Cross-session balanced accuracy fell to 55.4%, only slightly above the 50% random-guessing baseline
- Sensitivity dropped to 11.2%, indicating difficulty identifying samples written by the same person across sessions
- Validation performance may overestimate real-world reliability because validation data did not fully represent session variability

Siamese model limitations

- Sequential hyperparameter tuning may have overlooked interactions between parameters
- The final model performed worse than the tuning models, suggesting the selected combination may not have been optimal
- Repeated tuning on the same validation set may have produced overly optimistic performance estimates
- Differences in session representation across training, validation and test datasets complicated generalisation

Conclusion

Conclusion

- Both models struggled with cross-session handwriting variation
- The CNN could identify writers above chance, but its accuracy was not sufficiently reliable
- The Siamese network distinguished same-session pairs more effectively than cross-session pairs
- Variation in an individual's handwriting across sessions appears to be the main obstacle to reliable writer recognition

Thank you